Question ID: 99ea3715

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

If the graph of $27x + 33y = 297$ is shifted down 5 units in the *xy*-plane, what is the *y*-intercept of the resulting graph?

Answer

A. $(0, 4)$

B. $(0, 6)$

C. $(0, 14)$

D. $(0, 28)$

Correct Answer: A

Rationale

Choice A is correct. When the graph of an equation in the form $Ax + By = C$, where A, B, and C are constants, is shifted down k units in the *xy*-plane, the resulting graph can be represented by the equation $Ax + B(y + k) = C$. It's given that the graph of $27x + 33y = 297$ is shifted down 5 units in the *xy*-plane. Therefore, the resulting graph can be represented by the equation $27x + 33(y + 5) = 297$, or $27x + 33y + 165 = 297$. Subtracting 165 from both sides of this equation yields $27x + 33y = 132$. The *y*-intercept of the graph of an equation in the *xy*-plane is the point where the line intersects the *y*-axis, represented by the point $(0, y)$. Substituting 0 for x in the equation $27x + 33y = 132$ yields $27(0) + 33y = 132$, or $33y = 132$. Dividing both sides of this equation by 33 yields $y = 4$. Therefore, if the graph of $27x + 33y = 297$ is shifted down 5 units, the *y*-intercept of the resulting graph is $(0, 4)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the *y*-intercept of the graph of $27x + 33y = 297$ shifted up, not down, 5 units.

Choice D is incorrect and may result from conceptual or calculation errors.